

Infrastructure Governance & Compliance Manual

This technical manual documents the deterministic thresholds and security logic integrated into the ShineGuard Hulo platform, aligned with international urban engineering and cybersecurity standards.

Master Governance Summary

Category	Metric	Rule	Thesis-Level Justification	
Physics		Voltage	< 180V	Prevents SMPS current spikes & thermal runaway per IEC 60038.
Hardware		Thermal	> 85°C	Industrial casing threshold; prevents catastrophic silicon aging.
Safety		Current	> 5.0A	High-speed fire safety & circuit protection for node drivers.
Audit		Integrity	SHA-256	Non-Repudiation: Proves logs are mathematically untampered.

1. Electrical & Telemetry Thresholds

Metric	Nominal Range	Warning Level	Critical Trip	Compliance Standard
Voltage	210V - 240V	< 200V / > 250V	< 180V	IEC 60038
Temperature	20°C - 55°C	> 65°C	> 85°C	Industrial Grade
Current (A)	0.01A - 2.0A	> 2.5A	> 5.0A	IEC 60598-1
Ambient Light	10 - 1000 lux	< 10 lux	0 lux	EN 13201

Project Operational Workflows

1. The Telemetry & Alerting Engine Data is ingested from physical nodes in real-time. The infrastructure compares telemetry against deterministic thresholds. If a violation occurs, a high-severity alert is cascaded to both MySQL and Firebase layers.

2. Maintenance Governance Loop Administrators convert high-priority alerts into formal Work Orders. These are assigned to technicians whose resolution velocity (MTTR) is tracked for workforce efficiency auditing.

3. The Forensic Cryptographic Chain Every system operation is captured by the Audit Service. Using **HMAC-SHA256 Chaining**, each log entry is mathematically linked to the previous one, ensuring data-driven "Non-Repudiation."

System Architecture & Data Map

Diagram Legend:

- **Infrastructure Layer:** Physical streetlights and sensors generating raw telemetry.
- **Cloud Backend:** Deterministic processing and real-time database orchestration.
- **User Interface:** Live dashboard syncing via Firebase for immediate human oversight.
- **Forensic Security:** The HMAC-SHA256 chain providing data-driven legal accountability.

Deep-Security & Forensic Manifesto

ShineGuard Hulo implements a Multi-Layered Defense strategy, combining cryptographic integrity with administrative governance and active threat mitigation.

1. Active Threat Mitigation

- **Brute-Force Blacklisting (15-Min Lockout):** The system monitors failed login attempts. Upon reaching **5 consecutive failures**, the system triggers an automatic **15-Minute IP Lockout**. This mitigates automated dictionary and credential-stuffing attacks.
- **Velocity Control (Throttle):** Automated rate-limiting (60 req/min) mitigates automated scraping and low-level Denial-of-Service (DoS) attacks across all public endpoints.

2. Identity & Advanced Auth

- **Secure Session Authorization (SBA):** Bulk control operations are protected by a time-bound "5-Minute Authorization Window," requiring fresh re-authentication to prevent unauthorized access at administrative terminals.
- **Architectural MFA Ready:** The system infrastructure is built with dedicated hooks for Multi-Factor Authentication (MFA) integration, allowing for future-proofed TOTP (Google Authenticator) secondary verification.

3. Off-System Forensic Verification

- **The Manual Bypass & Audit Link:** Administrators can utilize the **Forensic Verification Link** (`tools/verify_logs.php`) to perform an independent, off-system check of the HMAC-SHA256 chain, verifying the authenticity of every city action log.

4. Cryptographic Integrity

- **HMAC-SHA256 Chaining:** Every log is chained to the previous entry, providing mathematical proof of **Non-Repudiation**. Even a database administrator cannot delete a single log without breaking the forensic chain.

-- END OF TOTAL SECURITY AUDIT --

Infrastructure Governance Status: HARDENED & AUDIT-READY (V4)